

Understanding the Capitalist Long Wave in Depth

Zhihu · 2026-09-03

Since the Industrial Revolution, periodic crises have been an inherent phenomenon of the capitalist world. In Marx's day a crisis broke out roughly every 7 to 10 years. This frequency changed only after the Second World War. Governments widely adopted Keynesianism and applied counter-cyclical measures to "smooth out" the cycle. These measures could not of course truly abolish crises, but they could indeed greatly reduce their impact, or postpone their outbreak — while also possibly lengthening the time needed for recovery. The Asian financial crisis of 1998, the American internet bubble crisis of 2000 and the global financial crisis of 2008 forced people once again to confront the reality that crises cannot be abolished.

Capitalist economic growth not only necessarily passes through the cycle of flourishing and decline; roughly every 40 to 50 years it produces a crisis of particularly great scale and particularly long recovery. Such crises have recurred throughout history: the crisis and long depression from the late 1830s to the early 1840s, the crisis and long depression of the 1870s, the global economic crisis and Great Depression of 1929, the stagflation crisis of the early 1970s, the global financial crisis of 2008 and the Great Recession that followed. Most researchers hold that we are still in the transitional phase of emerging from the long depression after the 2008 financial crisis.

These crises and the depressions that follow them are special not only because of the violence of the crisis and the length of the depression, but even more because of the reconstruction of the capitalist world order that the depression brings. After the depression of the 1840s, the capitalist world went through the revolutions of 1848, which cleared the road for capitalist development and enabled capital to expand rapidly within the various nation-states of the world between 1850 and 1870. The great depression of 1870 drove forward the concentration of capital and the formation of monopolies, intensified the colonial plunder of the capitalist countries, and prepared the conditions for imperialist war. The great depression at the end of the 1930s led to the Second World War, which at an exceptionally terrible cost completed the reconstruction of the superstructure of the capitalist world-system and established the US-led global order, preparing the political conditions for capitalism's global expansion. The stagflation of the 1970s drove the capitalist industrial countries to complete their neoliberal "reforms" and, through the globalisation of production, to reconstruct the division of labour and class relations on a world scale. And the long recession after 2008 produced the "anti-globalisation" wave in the centre countries of the capitalist world-system. Mankind seems to have returned to the state of imperial rivalry before the First World War. But this is only an illusion: the reconstruction of the political superstructure is under way, and a new multipolar order headed by China is very probably taking shape, preparing the conditions for a new round of capital's global expansion.

Why does the capitalist world experience such violent crises roughly every 40 years? More than a hundred years ago the Soviet economist Kondratieff, on the basis of a long-run study of wholesale price levels in the major capitalist countries, proposed that capitalist economic development contains a long cycle of about 50 years. The length of this cycle corresponds exactly to the interval between two great depressions in history. In fact, although the Kondratieff long wave is quite famous on the Chinese internet, where self-media articles about the "Kondratieff cycle" are often seen, most economists — including Marxist economists — do not acknowledge that it exists. Why?

First, the long wave is merely a generalisation based on empirical phenomena, and it is hard to explain the mechanism that produces it at the level of economic theory. Many Marxist economists, especially Trotskyist scholars, hold that although the turn of the long wave from prosperity to decline can be explained by capitalism's internal mechanisms, emerging from the depression requires political factors such as class struggle to drive it. And political factors are, for the internal working of the economy, exogenous.

Second, from the empirical side: Kondratieff discovered the long wave while analysing cyclical movements in commodity prices, but this cyclical fluctuation of prices disappeared after the 1940s.

On the disappearance of the volatility of the wholesale price index, the Marxist economist Anwar Shaikh, in his book *Capitalism: Competition, Conflict, Crises*, assembled several charts that vividly illustrate this change. As the charts below show, before the 1940s a fluctuation cycle of 40 to 50 years did indeed exist, whereas after the 1940s prices rose all the way without fluctuation.

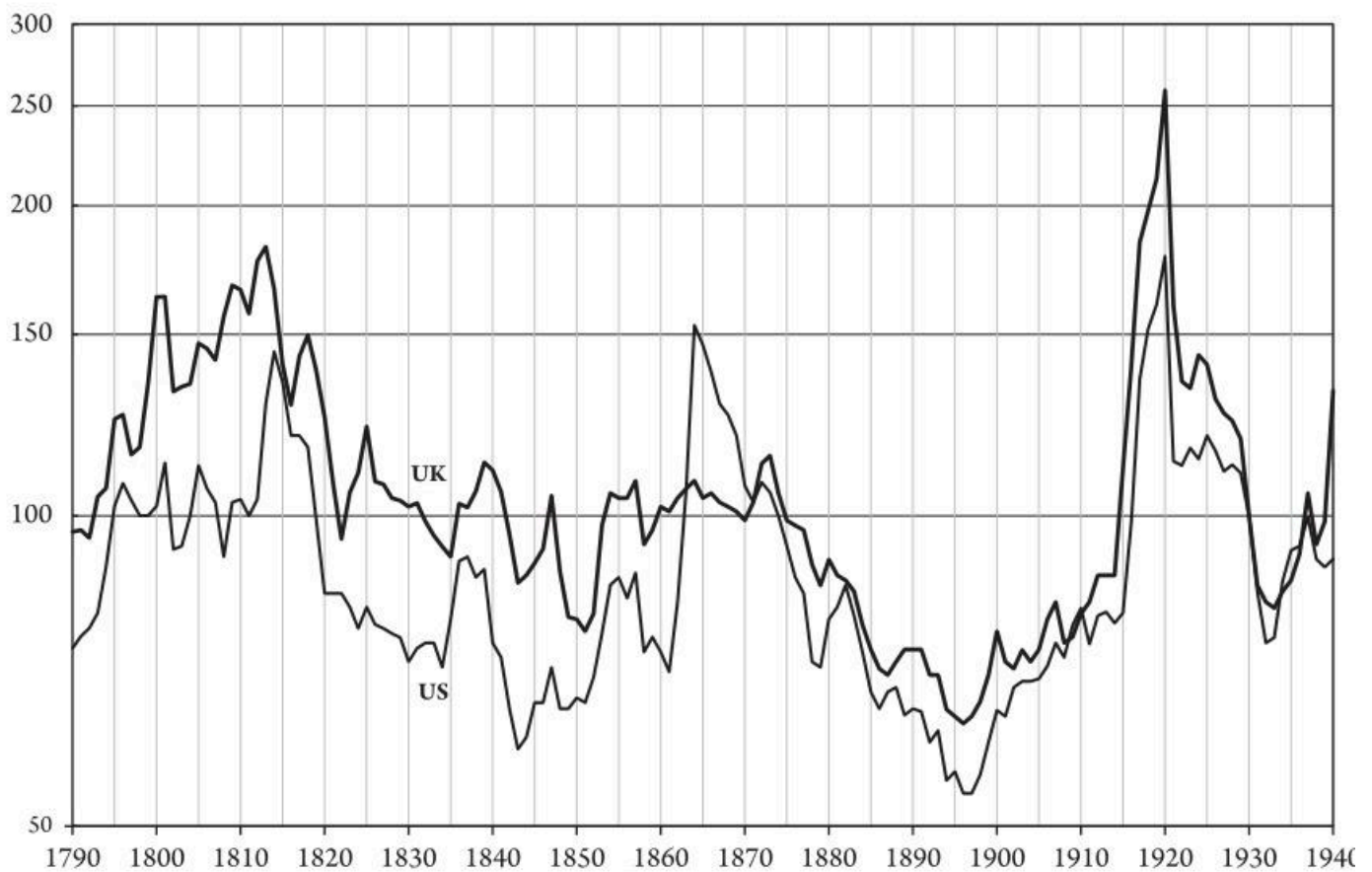


Figure 5.3 US and UK Wholesale Price Indexes, 1790–1940 (Log Scale, 1930 = 100)

Figure 5.3 shows the wholesale price indices of the two principal countries of the capitalist world (the United States and Britain) from 1790 to 1940, the same material presented earlier in Figure 2.9 of Chapter 2. Three points deserve note: the movements of the two countries' price indices are on the whole very similar; these indices contain long-period fluctuations, from which long wave theory is precisely derived (van Duijn 1983, Chapter 5); and a striking fact — over the whole 150-year period these price indices display no long-term upward or downward trend. Indeed, Jastram (1977, p. 189) points out that in Britain the purchasing power of gold "in the mid-twentieth century, astonishingly, remained remarkably consistent with that of the mid-seventeenth century."

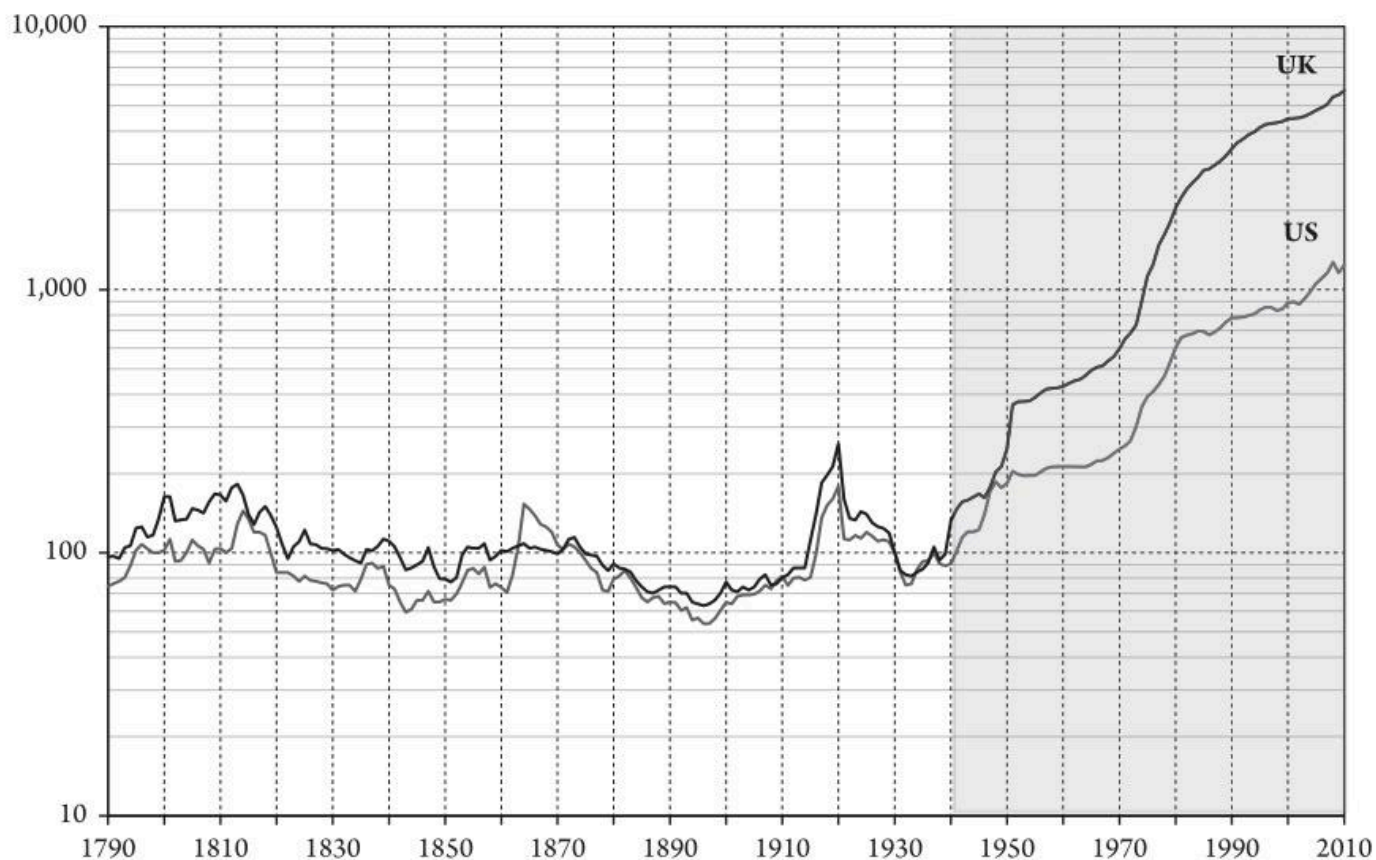
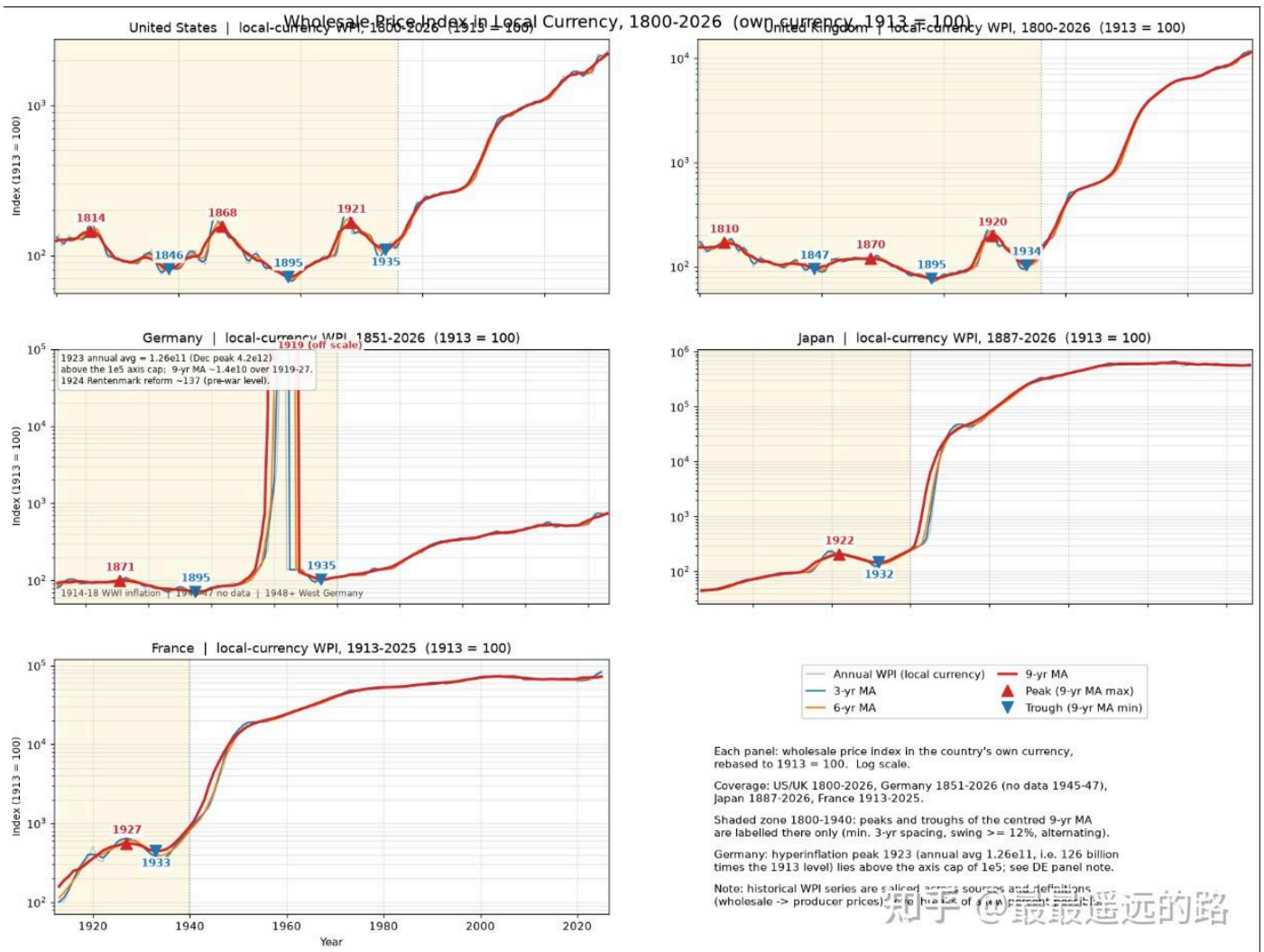


Figure 5.4 US and UK Wholesale Price Indexes, 1790–2010 (Log Scale, 1930 = 100)

Figure 5.4 extends the time span to 2010 and shows that these patterns changed drastically after 1939/40. In this new period prices display a sustained upward trend; the relatively steady fluctuations of the earlier price level have been drowned by the cumulative effect of continuous inflation. By 2010 the price level in Britain stood 58 times above its 1939 pre-war benchmark, while that of the United States stood 14 times above its 1940 benchmark.

The same situation can be seen in the charts the author has drawn from economic data obtained for several major capitalist countries. After 1940 no long wave can be seen in them.



Drawn by the author: https://github.com/sacanlee/long_wave_ode/blob/main/figures/wpi_chart_local_currency.png

But Anwar Shaikh's research shows that if commodity prices are converted from pounds and dollars into gold, the cycle can still be observed after 1940. Moreover, in the early 2000s Shaikh accurately predicted, on the basis of the cycle's regularities, that a great crisis of the downswing would occur around 2007.

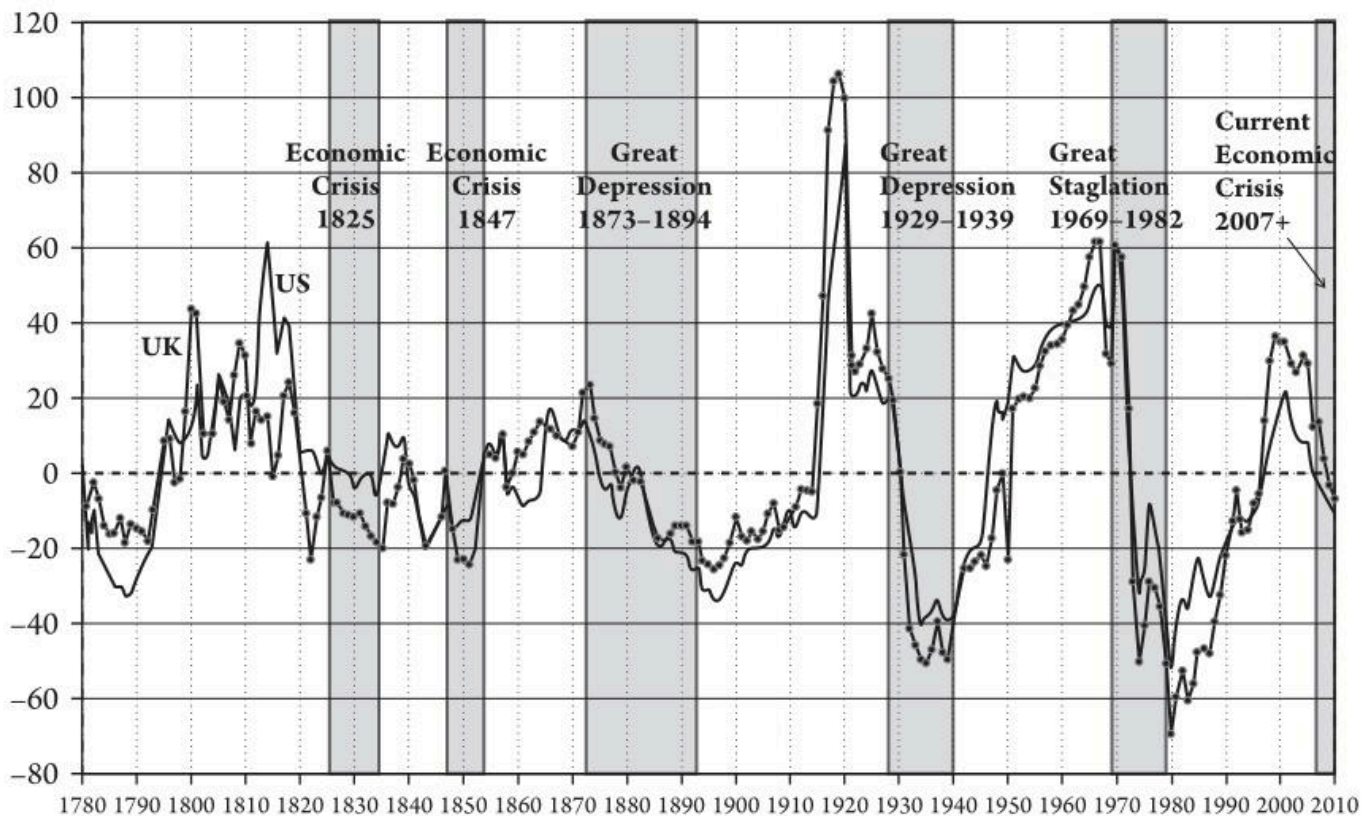


Figure 16.1 US and UK Golden Waves, 1786–2010 (1930 = 100) Deviations from Cubic Time Trends Source: Appendix 5.3 data tables.

Figure 16.1 shows the movement of commodity prices in the United States and Britain expressed in gold (after detrending) between 1790 and 2010. It also marks the points at which the various general crises and great depressions occurred; these crises can be seen to begin usually in the middle of the long cycle's downswing. The reader will note that the 2007 financial crisis came, one might say, right on schedule. Every general crisis originated in the developed countries, although as globalisation advanced more and more developing countries were drawn in. The HP-filtered data reveal two clear long cycles since 1897: 1897 to 1939 (42 years) and 1939 to 1983 (44 years), both measured trough to trough. A general crisis usually breaks out eight or nine years after each cycle's peak and lasts until about eighteen years after the peak. Since 2003, in my classes and public lectures, combining the 2000 peak already visible at the time with the average cycle characteristics, I have predicted that the next crisis would begin in 2008–2009 and last until 2018. — Anwar Shaikh, *Capitalism: Competition, Conflict, Crises*

Other economists, following Anwar Shaikh's method, have analysed the movement of US commodity prices denominated in gold and likewise found a long wave cycle of 40 to 50 years. We are at present in the transition from the downswing of the fifth cycle to the upswing of the sixth.

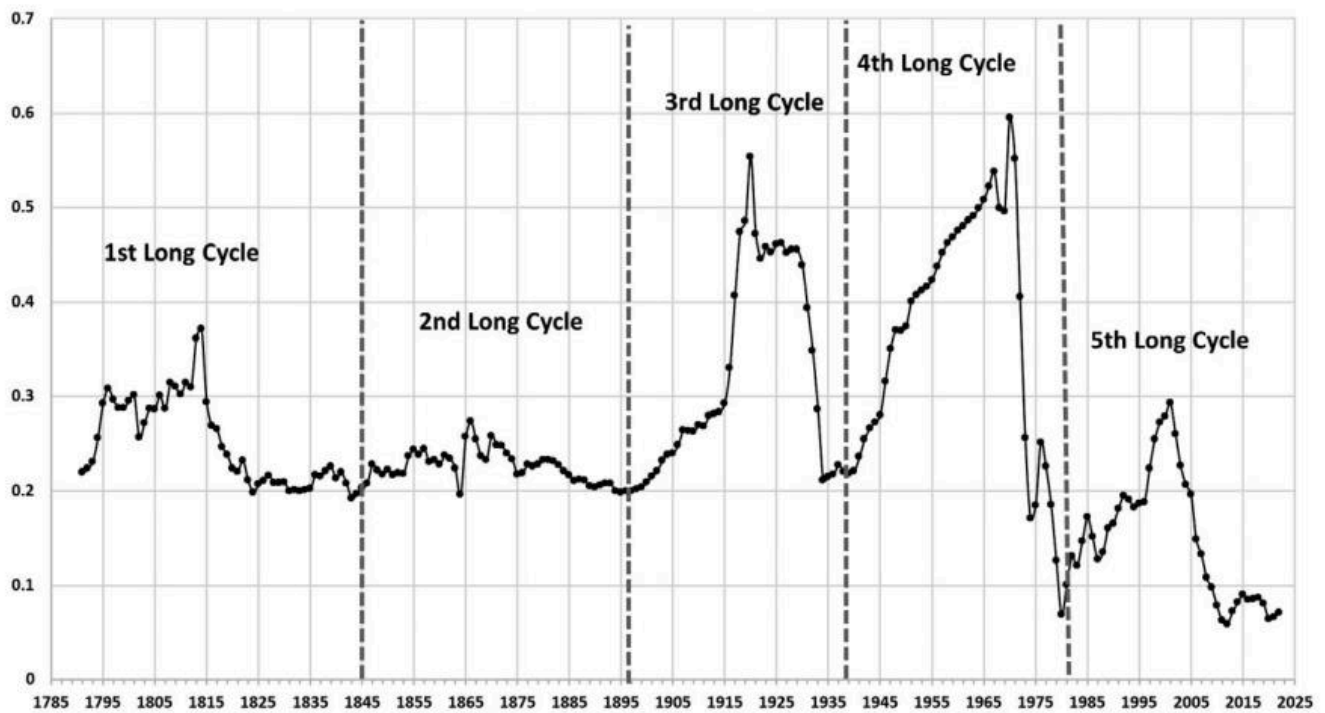
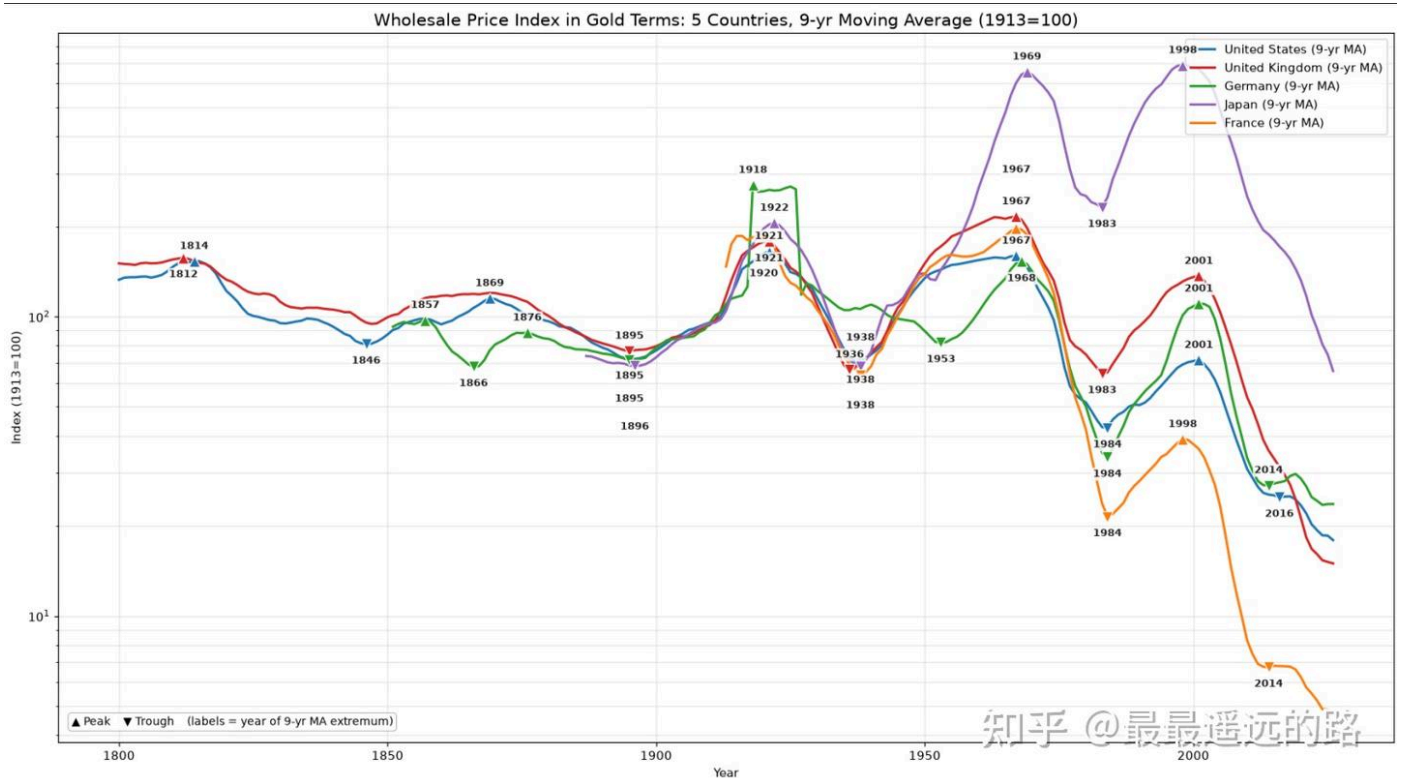


Figure 5.1 Evolution of 'golden price', US economy 1790–2022 (2012 base year).

Following the method of Kondratieff, we analyse using data whose span is long enough to support the long wave hypothesis (covering wholesale prices and the price of gold). Specifically, we construct the ratio of the US GDP deflator (which is no different from the wholesale price index) to the index of the gold price for the period 1790 to 2022 (see Figure 5.1). The ratio of the wholesale price index to the price of gold constitutes the "golden price", representing the quantity of gold that one unit of output can purchase. Both prices are expressed as indices with 2012 as the base period. Figure 5.1 shows that the cyclical fluctuations in the wholesale price of gold (that is, the "golden price") reflect the long wave divisions listed in Table 5.1. The cyclical character of the US economy is thus clearly revealed; given the interconnections among national economies (and these connections are growing ever stronger), this cyclicity of the US economy has also dominated the course of the international economy. The discussion above shows that although the evolution of the capitalist economy is full of turbulence, its time-series data do indeed contain a certain cyclical regularity. Figure 5.1 shows the five long cycles that the American capitalist economy has experienced in terms of the "golden price" since the (first) Industrial Revolution. — Nikolaos Chatzarakis, Persefoni Tsaliki, Lefteris Tsoulfidis, *Economic Growth and Long Cycles: A Classical Political Economy Approach*

To confirm the reality of the long wave's existence, the author obtained WPI data and gold price data for the major capitalist countries and drew the following wholesale price index denominated in gold. It can be seen that the capitalist world has indeed passed through five long wave cycles; the fifth began around 1983, peaked around 2000, and by 2026 has still not emerged from its downswing.



Drawn by the author: https://github.com/sacanee/long_wave_ode/blob/main/figures/wpi_chart_combined_9yMA_log.png

So far as empirical data go, then, the long wave still exists. In fact Shaikh also points out that Kondratieff published two kinds of data in his book: the first denominated in money, and another denominated in gold. Kondratieff's charts, however, were drawn from the money-denominated data, while the gold-denominated data were presented only in tables. Since only the charts are well known, people continued to study the money-denominated price data and found that the long wave had vanished. Shaikh's long research on the Kondratieff long wave has brought this valuable economic concept back into the mainstream of Marxist economic research.

But even if we accept that the long wave exists, what is the principle that produces it? Why is the long wave 40 to 50 years long? This remains a puzzle.

The key to understanding the long wave is to explain clearly two transitions: the transition from upswing to downswing, and the transition from downswing to the next upswing.

Why must the rapid expansion of the capitalist economy meet a crisis after 20 to 30 years? Marxist economists are broadly agreed on this question, holding that crises arise from a fall in the rate of profit, and that the fundamental factor in that fall is the rising organic composition of capital. Capitalist competition forces enterprises to keep increasing investment, raising labour productivity and increasing the share of constant capital in total capital, which leads to a continuous fall in the rate of profit. If labour remains in persistent oversupply, capitalists can maintain the rate of profit by depressing the value of labour power and raising the rate of exploitation. As the figure below shows, the real wages of American workers grew faster than labour productivity between the Second World War and the 1970s, which means that besides the fall in the rate of profit caused by the rising share of constant capital, factors such as the

exhaustion of the reserve army of labour, strong trade unions and social democratic parties were squeezing the rate of profit further. This brought about the transition of the fourth long wave from upswing to downswing.

After the 1980s, however, as workers of the Global South entered the capitalist labour market en masse, as American enterprises moved abroad and the trade unions were weakened, the growth rate of American workers' real wages fell far behind the growth rate of productivity. That is to say, not only did the fall in the share of constant capital brought by "deindustrialisation" push the rate of profit up; the rise in the rate of exploitation raised it further. Capitalism was thus able to emerge from the downswing of the fourth long wave and begin the upswing of the fifth.

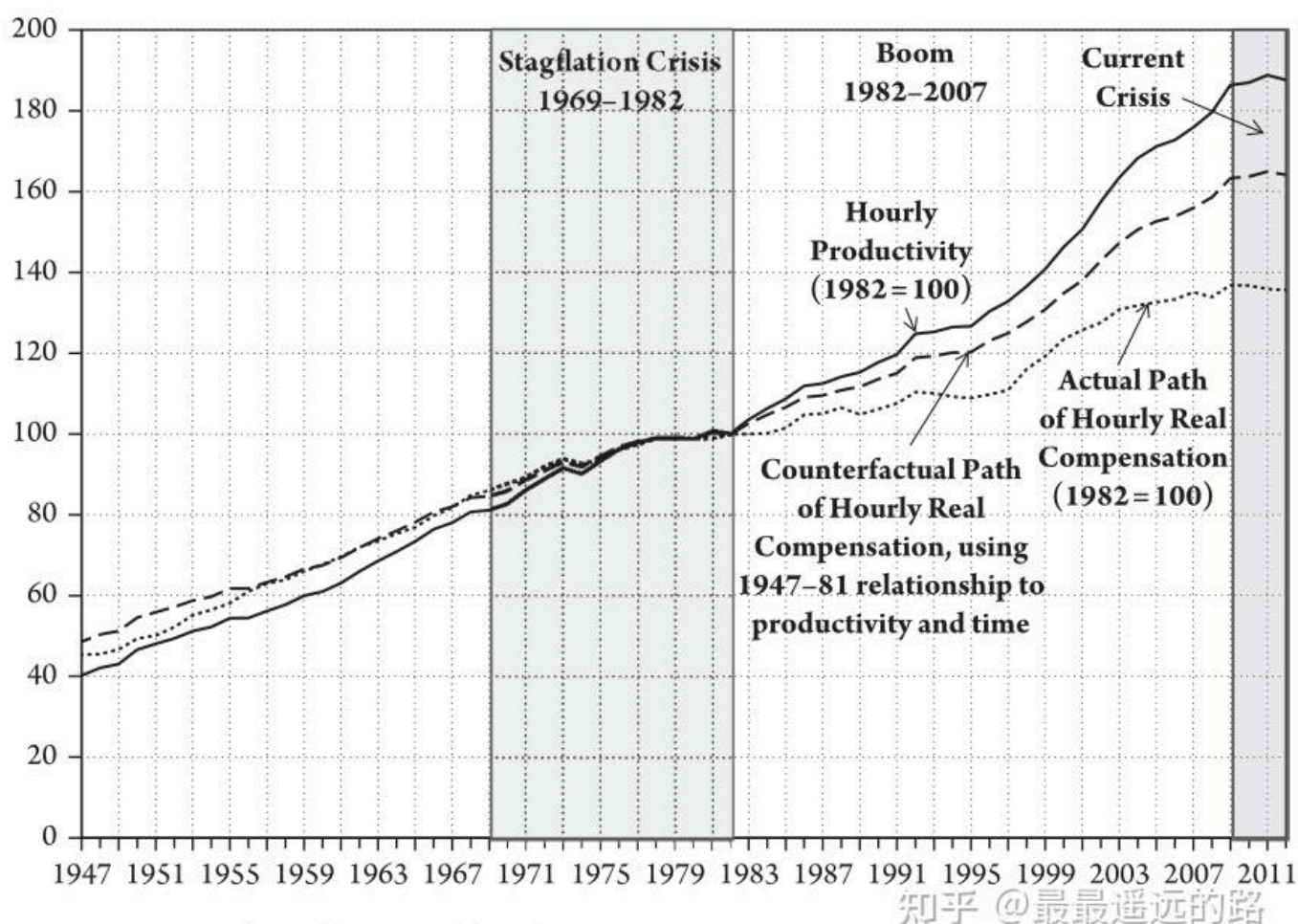


Figure 16.3 Hourly Real Wages and Productivity, US Business Sector, 1947–2012 (1982 = 100)

Figure 16.3 shows the relationship between output per hour and real compensation per hour (the real wage) in the US business sector from 1947 to 2008. Even under normal conditions real wages usually grow more slowly than productivity; however, from the Reagan years of the 1980s onward the growth of real wages slowed markedly. The difference becomes obvious if one compares the level of real wages since 1980 with the level they would have reached had the post-war relationship with productivity growth been maintained. This deviation from the historical trend stemmed from a two-pronged strategy: on the one hand the suppression of the public and private institutions that supported labour, on the other the rise of

globalisation, which brought the world's vast cheap labour force into more direct competition with the labour markets of the developed countries.

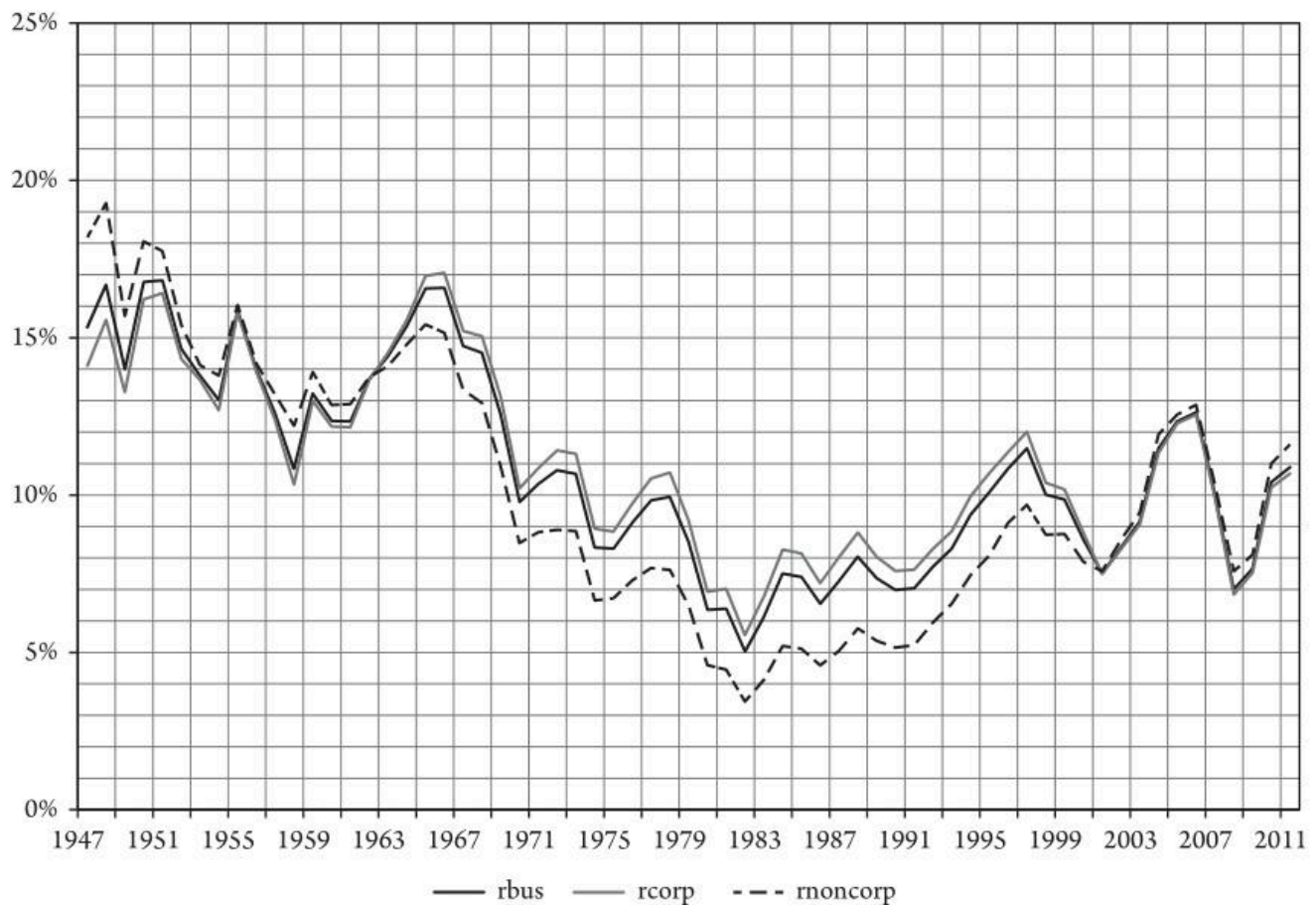


Figure 6.1 Corporate and Non-Corporate Profit Rates

The rate of profit of US enterprises

In the transition of the long wave from upswing to downswing, therefore, we see two factors:

1. The bourgeoisie's constant replacement of people by machines raises the organic composition of capital. This is an internal factor in the working of the economy.
2. Class struggle (trade unions, left-wing forces at home, pressure from the Soviet camp, and so on) pushes wages up, which in turn gives rise to a profit squeeze.

And in the transition from the downswing of one long wave to the upswing of the next, we likewise see factors that raise the rate of profit:

1. The destruction of vast quantities of capital during the depression, and the acceleration of technological innovation, reduce the total value of the existing capital stock. This is an internal factor in the working of the economy.

2. The bourgeoisie defeats the working class by suppressing trade unions and bringing in immigrants, rebuilding a firm order of exploitation and raising the rate of exploitation.
3. A low rate of profit forces capitalists to invest in research and development, so innovation is faster in depressions than in booms.

How do these factors operate in the rise and fall of the long wave cycle? Nikolaos Chatzarakis, Persefoni Tsaliki, Lefteris Tsoulfidis and others attempt to solve this problem by mathematical modelling in their recently published book *Economic Growth and Long Cycles: A Classical Political Economy Approach*. Proceeding from the basic principles of Marxist political economy, they derive 5 differential equations; solving these equations reveals the secret of the long wave.

$$\text{Profitability: } \dot{r} = -a_1(s_C - s_V)r^2 + a_2(\delta + \tau - s_V)r$$

$$\text{Investment in variable capital: } \dot{s}_V = s_C r - \delta - \tau - n$$

$$\text{Investment in fixed capital: } \dot{s}_C = s_C(\tau - s_V r)$$

$$\text{Devaluation rate: } \dot{\delta} = b_0 + \delta(g_w - s_c r)$$

$$\text{Technological change: } \dot{\tau} = b_1 \text{Erf}(\delta - b_2)$$

For the specific meaning, see Chapter 8 of *Economic Growth and Long Cycles: A Classical Political Economy Approach*. For the derivation, see the book itself.

As the figure below shows, the numerical solution of the five differential equations takes the form of a cycle of about 50 years, which corresponds roughly to the historical long wave cycles.

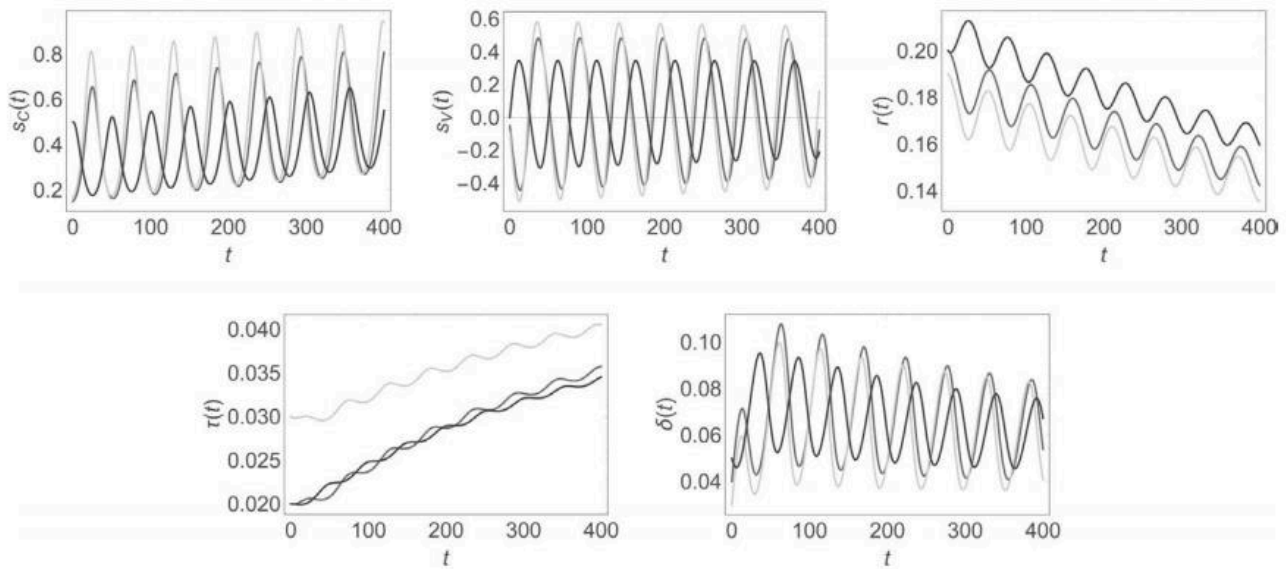
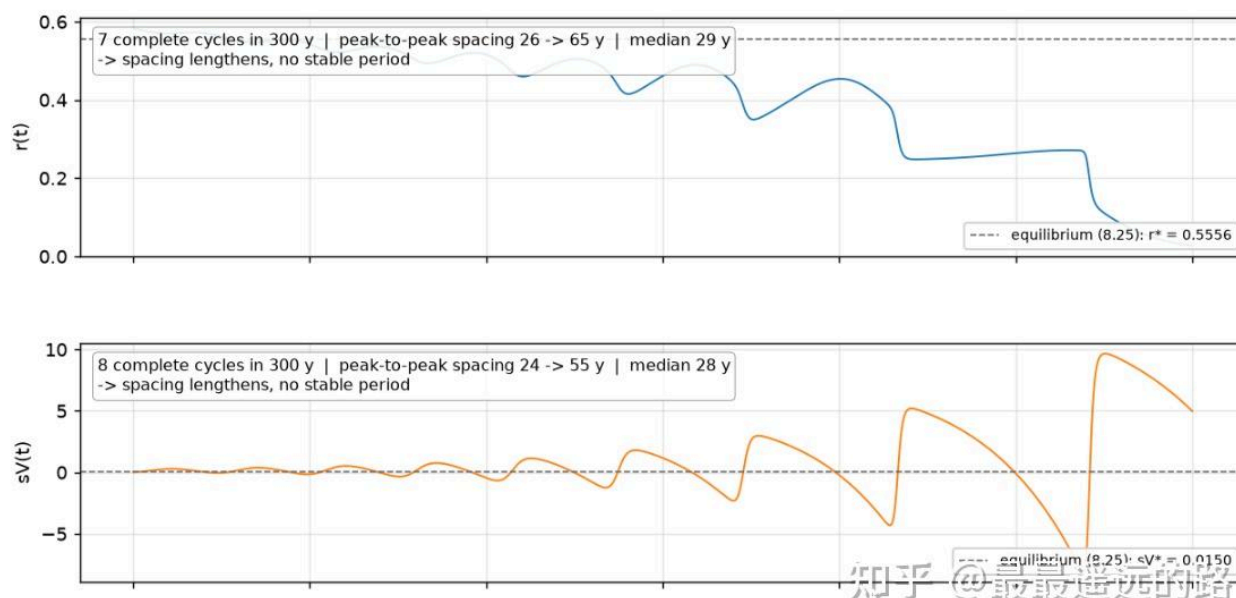


Figure 8.13 Numerical solutions of differential Equations (8.15), (8.16), (8.17), (8.18) and (8.19). 知乎 @最遥远的路

Source: *Economic Growth and Long Cycles: A Classical Political Economy Approach*, Chapter 8

The author has also tested the differential equations of *Economic Growth and Long Cycles*. In the course of testing, some problems in the solution process came to light (see: https://github.com/sacanlee/long_wave_ode/blob/main/docs/Technical_Report_Ch8_Model_Errata.docx), and these have been reported to the authors, with discussion still under way. The authors say that if a second edition is published, it may be updated. However, the author's own test over a span of 300 years shows that the cycle does indeed exist. This suggests that these differential equations very probably do express the underlying mechanism of the long wave phenomenon in the capitalist economy.

Nonlinear solution of the long-wave model (8.15)-(8.19) at the book parameters
 $a_1=0.04, a_2=0.01, b_0=0.004, b_1=0.005, b_2=0.05, gw=0.03, n=-0.015$ | initial values = closed-form equilibrium (8.25) $\times 1.05$, wind



Drawn by the author; full figure at:

https://github.com/sacanlee/long_wave_ode/blob/main/figures/ode_solution_book_params.png

The author adjusted the value of n , the population growth rate, and tested the cycle against historical data: the model's predicted cycles for the nineteenth century are much shorter than the actual ones, but the predictions for the twentieth century are relatively accurate.

3. Script predictions (history_compare.py, book parameter values)

For each wave, input the full-wave n (working-age 15–64) (other parameters $a_1=0.04, a_2=0.01, b_0=0.004, b_1=0.005, b_2=0.05, gw=0.03$), compute closed-form equilibrium $\rightarrow$ analytical Jacobian $\rightarrow$ dominant eigenvalue $\rightarrow$ period; then measure rise/fall segments for a 5% perturbation, both linear and nonlinear (multi-cycle, sC measure).

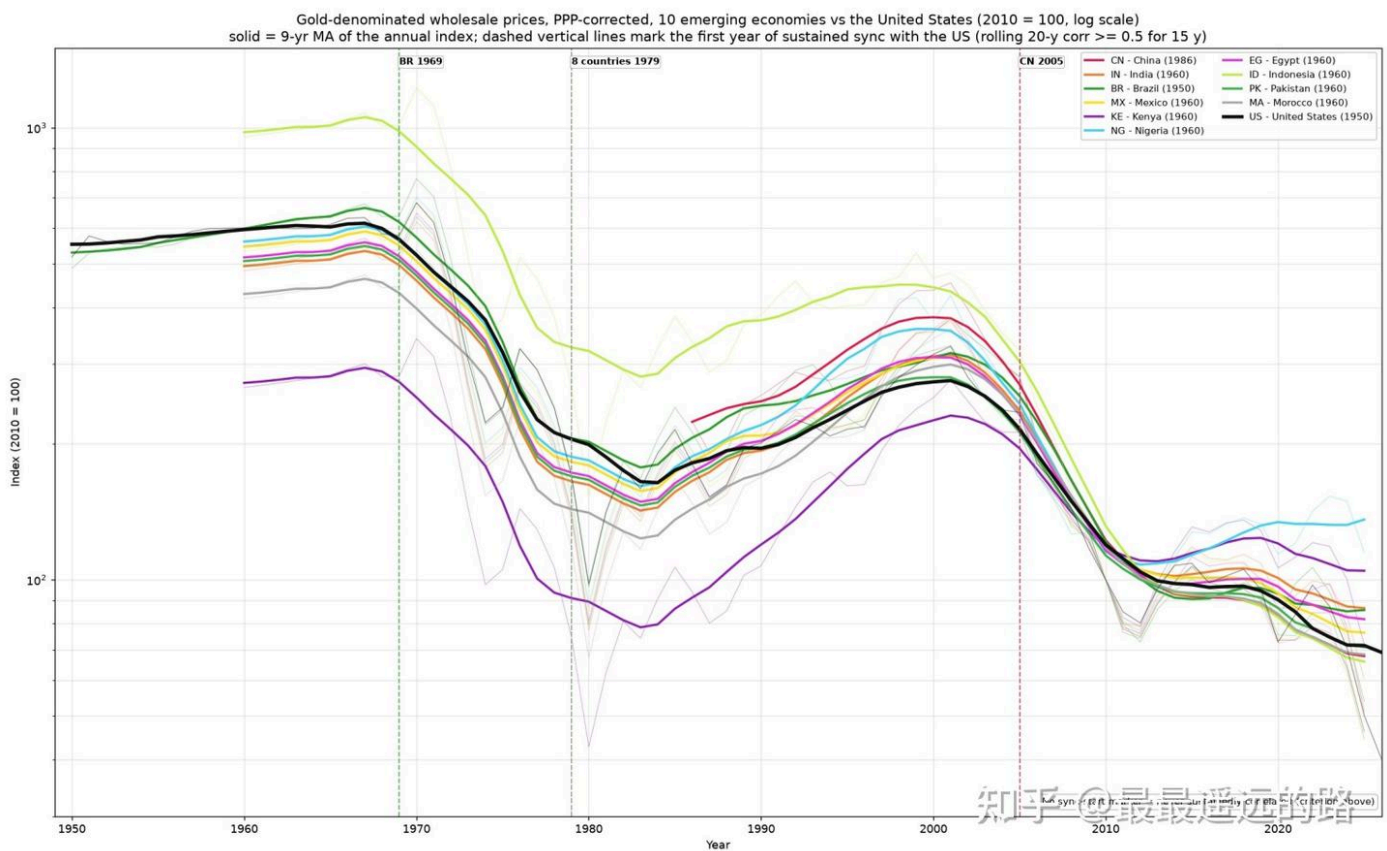
Historical wave	n (15–64)	Actual T	Predicted T (linear)	Bias	Model rise/fall (linear)	Model rise/fall (nonlinear, sC)	Actual rise/fall
W1 Industrial Rev.	0.537%	55 y	35.7 y	–35%	17.7 / 17.9	20.4 (50%) / 20.1	25 / 30
W2 Victorian	0.493%	51 y	35.4 y	–31%	17.6 / 17.8	20.2 (50%) / 19.9	28 / 23
W3 Belle Époque	0.907%	44 y	37.9 y	–14%	18.8 / 19.0	22.6 (51%) / 21.5	24 / 20
W4 Golden Age	1.691%	42 y	43.1 y	+3%	21.3 / 21.8	29.3 (55%) / 24.2	26 / 16
W5 Information	1.623%	40 y	42.6 y	+7%	21.1 / 21.5	20.5 (51%) / 24.3	26 / 15

Reference: https://github.com/sacanlee/long_wave_ode/blob/main/docs/history_comparison.md

On the whole this model does provide a strong underlying mechanism for the long wave cycle, and raises the scientific standing of long wave theory.

If the long wave is an inherent mechanism of capitalism, then two questions need answering. First, is the long wave a phenomenon only of the centre countries of the world-system, or has it already spread across the globe? Second, when will the next long wave begin, and when will it end?

The figure below shows the movement of the wholesale price index in China and India and in other developing countries, including some typical African countries. Because the currencies of developing countries have depreciated heavily, the data were processed using purchasing power parity. The resulting wholesale price index in gold also displays a cyclicality broadly in step with America's. This shows that the long wave cycle is a phenomenon of the capitalist world-system as a whole, acting on all the countries within the world economy. The more industrialised a developing country is, the more closely its cycle is in step with America's. China has also shown basically the same trend since the 1980s.



Source: https://github.com/sacanlee/long_wave_ode/blob/main/figures/wpi_em_gold_denominated_sync.jpg

As for when the sixth and seventh long waves begin and end, that is at present very hard to determine and can only be conjectured. Since we are now in the final stage of the downswing of the fifth long wave, and since an AI bubble makes a crisis very likely — though the crisis is very likely to be followed by a rapid rebound — we assume that the sixth long wave begins in 2030. Of the factors influencing the duration of a long wave, the key variable we can identify is the rate of population growth. Taking the medium scenario of the UN's population projections, projecting the growth rate of the labour force and feeding it into the differential equations yields the following result: the cycle begins in 2030, peaks around 2048 and ends in

2068. On historical experience a crisis breaks out about 7 years after the peak, so the next great crisis will fall at roughly 2055.

Segment	Period	n (CAGR, 15–64)	Model T (eigenvalue)	Nonlinear sC: T	Rise share
Upswing	2030–2045	+0.49%	35.4 y	39.2 y	50.3%
Downswing	2045–2070	+0.16%	33.5 y	37.6 y	50.2% (fall 49.8%)

Wave 6 timeline (main measure: eigenvalue period + nonlinear rise share):

Event	Time	Computation
Trough (start)	2030	scenario premise (tail of the Wave 5 crisis)
Peak (boom top / crisis onset)	≈ 2048–2050	2030 + rise ≈ 2030 + 0.503×35.4 (≈17.8 y) → 2048; with the nonlinear T ≈ 2050
Trough (crisis bottom)	≈ 2065–2068	peak + fall (≈0.498×33.5 ≈ 16.7 y; nonlinear-T measure 18.7 y)
Wave 6 total period	≈ 34.5–38.4 y	17.8 + 16.7 (eigenvalue measure) ≈ 34.5; nonlinear sC ≈ 38.4
Next trough	≈ 2065–2068	closure of the full long wave from 2030

Source: https://github.com/sacanlee/long_wave_ode/blob/main/docs/forecast_2050_2080.md

Calculating the seventh long wave in the same way gives a cycle running roughly from 2068 to 2102, with its high point around 2086 and a great crisis in the early 2090s.

Wave 7 phase n (15–64, UN medium data): 2070–2085 = −0.137%/yr (upswing segment), 2085–2100 = −0.103%/yr (downswing segment); full wave (2070–2100) −0.120%/yr. 15–64 declines gently from 6.286bn (2070) to 6.064bn (2100) — much milder than the deep scenarios (global population inertia buffers).

Model	Rise (share)	Peak	Fall	Total period	Trough
Eigenvalue measure	15.8 y (49.7%)	≈2086	16.1 y	32.0 y	≈2102
Nonlinear sC measure	17.4 y (49.7%)	≈2087	17.8 y	35.2 y	≈2105

Source: https://github.com/sacanlee/long_wave_ode/blob/main/docs/forecast_2050_2080.md

The 25 years from 2030 to 2055 will be a period of rapid capitalist development. In the 30 years beginning now, the great majority of the Southern regions, including Africa and South Asia, will complete their industrialisation, and the overwhelming majority of the labouring people of mankind will become proletarians. At the same time the economies of the various nation-states are so highly integrated that the cost of war is enormous and the difficulty of launching a world war extremely great. This will be the most important feature distinguishing the crises and depressions of the 2050s and 2060s from those of the past. Because the slowdown of population growth, even turning into negative growth, has shortened the cycle,

this century will still see two great crises on the scale of the 2008 financial crisis. The explosion of class contradictions that these two crises bring, and their resolution, will decide the fate of mankind.

Appendix: for the relevant code, see: https://github.com/sacanlee/long_wave_ode

Core script: https://github.com/sacanlee/long_wave_ode/blob/main/scripts/long_wave_model.py

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